

Problem

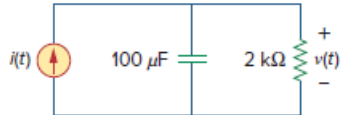
For the circuit in Fig. 17.80,

$$i(t) = 20 + 16 \cos(10t + 45^\circ) + 12 \cos(20t - 60^\circ) \text{ mA}$$

(a) find $v(t)$, and

(b) calculate the average power dissipated in the resistor.

Figure 17.80:



Step-by-step solution

Step 1 of 7

(a)

The dc component of the input current is

$$I_{dc} = 20 \text{ mA}$$

The AC components are

$$I_{\omega=10 \text{ rad/s}} = 16 \angle 45^\circ \text{ mA and}$$

$$I_{\omega=20 \text{ rad/s}} = 12 \angle -60^\circ \text{ mA}$$

The frequency domain equivalent parameters are,

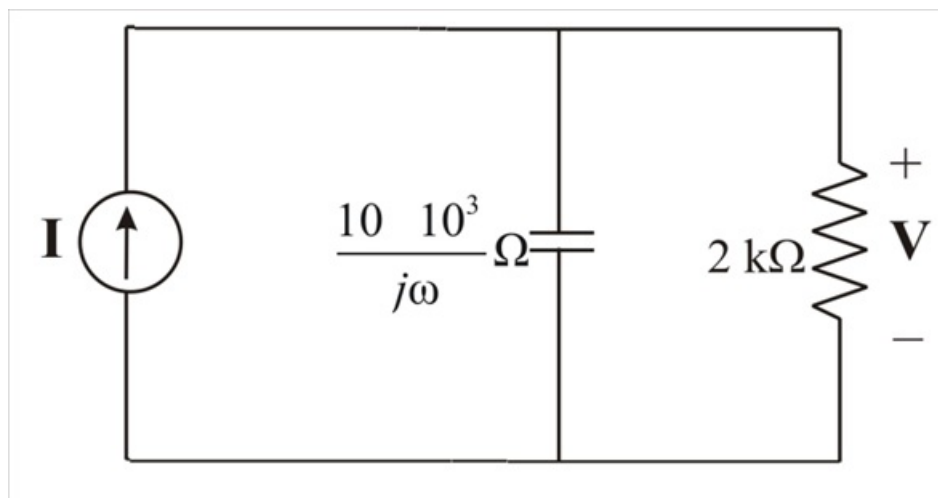
$$i(t) \Rightarrow I$$

$$v(t) \Rightarrow V$$

$$100 \mu\text{F} \Rightarrow \frac{1}{j\omega(100 \times 10^{-6})}$$

$$100 \mu\text{F} \Rightarrow \frac{10 \times 10^3}{j\omega} \Omega$$

Thus the frequency domain circuit is,



Step 2 of 7

Applying Ohm's law the voltage V is,

$$\begin{aligned} \mathbf{V} &= \mathbf{I} \left[\frac{10 \times 10^3}{j\omega} \parallel 2 \times 10^3 \right] \\ &= \mathbf{I} \left[\frac{\left(\frac{10 \times 10^3}{j\omega} \right) (2 \times 10^3)}{\left(\frac{10 \times 10^3}{j\omega} \right) + (2 \times 10^3)} \right] \end{aligned}$$

Simplify further.

$$\begin{aligned} \mathbf{V} &= \mathbf{I} \left[\frac{(10 \times 10^6)(2 \times 10^3)}{(10 \times 10^3) + (2 \times 10^3)j\omega} \right] \\ &= \mathbf{I} \left[\frac{20 \times 10^3}{10 + j2\omega} \right] \\ &= \mathbf{I} \left[\frac{10 \times 10^3}{5 + j\omega} \right] \dots\dots (1) \end{aligned}$$

Step 3 of 7

For the dc component, $\omega = 0$.

Thus the equation (1) is given by

$$\begin{aligned} \mathbf{V}_{dc} &= \mathbf{I}_{dc} \left[\frac{10 \times 10^3}{5 + j(0)} \right] \\ &= (20 \times 10^{-3}) \left[\frac{10 \times 10^3}{5} \right] \end{aligned}$$

$$\mathbf{V}_{dc} = 40$$

Therefore, the dc component voltage is $\mathbf{V}_{dc} = 40 \text{ V}$.

Step 4 of 7

For $\omega = 10 \text{ rad/s}$ equation (1) is,

$$\begin{aligned} \mathbf{V}_{\omega=10 \text{ rad/s}} &= \mathbf{I}_{\omega=10 \text{ rad/s}} \left[\frac{10 \times 10^3}{5 + j(10)} \right] \\ &= (16 \times 10^{-3} \angle 45^\circ) \left[\frac{10 \times 10^3}{5 + j10} \right] \\ &= \frac{160 \angle 45^\circ}{11.18 \angle 63.44^\circ} \\ \mathbf{V}_{\omega=10 \text{ rad/s}} &= 14.31 \angle -18.44^\circ \end{aligned}$$

Therefore, for $\omega = 10 \text{ rad/s}$ the output voltage is $\mathbf{V}_{\omega=10 \text{ rad/s}} = 14.31 \angle -18.44^\circ \text{ V}$.

Step 5 of 7

For $\omega = 20 \text{ rad/s}$ equation (1) becomes:

$$\begin{aligned} \mathbf{V}_{\omega=20\text{rad/s}} &= \mathbf{I}_{\omega=10\text{rad/s}} \left[\frac{10 \times 10^3}{5 + j(20)} \right] \\ &= (12 \times 10^{-3} \angle -60^\circ) \left[\frac{10 \times 10^3}{5 + j20} \right] \\ &= \frac{120 \angle -60^\circ}{20.62 \angle 75.96^\circ} \\ \mathbf{V}_{\omega=20\text{rad/s}} &= 5.82 \angle -135.96^\circ \end{aligned}$$

Therefore, for $\omega = 20 \text{ rad/s}$ the output voltage is $\mathbf{V}_{\omega=20\text{rad/s}} = 5.82 \angle -135.96^\circ \text{ V}$.

Step 6 of 7

The complete output voltage in frequency domain is,

$$\begin{aligned} \mathbf{V} &= \mathbf{V}_{\text{dc}} + \mathbf{V}_{\omega=10\text{rad/s}} + \mathbf{V}_{\omega=20\text{rad/s}} \\ \mathbf{V} &= 40 + 14.31 \angle -18.44^\circ + 5.82 \angle -135.96^\circ \end{aligned}$$

The output voltage $v(t)$ in time domain is,

$$40 + 14.31 \cos(10t - 18.44^\circ) + 5.82 \cos(20t - 135.96^\circ) \text{ V}$$

Step 7 of 7

(b)

The average power dissipated in the resistor is,

$$\begin{aligned} P &= V_{\text{dc}} I_{\text{dc}} + \frac{1}{2} \sum_{n=1}^{\infty} V_n I_n \cos(\theta_n - \phi_n) \\ &= (40)(20) + \left[\frac{1}{2} (14.31)(16 \times 10^{-3}) \cos(18.44^\circ - (-45^\circ)) + \right. \\ &\quad \left. \frac{1}{2} (5.82)(12 \times 10^{-3}) \cos(135.96^\circ - 60^\circ) \right] \\ &= 800 + \left[(114.48 \times 10^{-3}) \cos(63.44^\circ) + \right. \\ &\quad \left. (34.92 \times 10^{-3}) \cos(75.96^\circ) \right] \end{aligned}$$

Simplify further.

$$\begin{aligned} P &= 800 + \left[(114.48 \times 10^{-3})(0.447) + \right. \\ &\quad \left. (34.92 \times 10^{-3})(0.243) \right] \\ &= 800 + [0.0512 + 0.0085] \\ &= 800.0597 \end{aligned}$$

Therefore, the average power P dissipated in the resistor is 800 W .